

Morality in infants

Helper/Hinderer Study

MATH 213

Introduction

The "Infant Cognition Center" is a lab at Yale University that studies the thinking of infants. A paper published in 2007 by researchers affiliated with this center ("Social Evaluation by Preverbal Infants" by Hamlin, Wynn, and Bloom) studied the question of whether 10-month old infants would prefer someone that helps another individual to someone that hinders another individual.

Introduction, continued

The following excerpt from the paper explains part of what the researchers did:

[I]nfants saw a character (the 'climber', made of wood and with large eyes glued onto it) initially at rest at the bottom of a hill. During a habituation phase, infants saw events in which the climber repeatedly attempted to climb the hill, and on the third attempt was either aided up by a helper who pushed it from behind, or was pushed down by a hinderer. Infants saw alternating helping and hindering trials with looking time measured on each trial, until their looking time reached a pre-set criterion indicating they had sufficiently processed these events... We asked here how infants ... evaluate helpers and hinderers. Will witnessing one individual's actions towards an unknown third party affect infants' attitude towards that individual? ...

[O]ur choice measure examined infants' attitudes towards the helper and hinderer. Infants were encouraged to choose between the two (that is, reach for one). Infants robustly chose the helper (14 of 16 10-month-olds).

Please watch the two short “habituation” videos at the following site to see what the researchers did:

<http://campuspress.yale.edu/infantlab/media/>

Evaluating the strength of the evidence

We would like to evaluate the strength of the evidence. A total skeptic looking at this study might suggest that the researchers just got lucky: the babies were equally likely to pick either the helper or hinderer, and it was just by chance that so many (14 out of 16 babies) picked the helper.

As a next step in analyzing the data, we will evaluate the skeptic's proposal using a model involving random chance that makes precise the skeptic's proposal. We will call this mathematical model the **null model**.

Our null and alternative hypotheses here are:

Null hypothesis, H_0 : The babies are equally likely to pick helper or hinderer. The study result of 14/16 occurred by chance alone.

Alternative hypothesis, H_a : The babies are more likely to pick helpers.

Your individual job

We imagine that each of the babies was making a pick at random that we can model with a coin flip. The choices of all 16 babies can be modeled with 16 coin flips. To see how well the skeptic's proposal can account for the data in the experiment, we will see how common it is for coin flipping to give 14 or more "heads" flips out of 16 trials.

Please flip a coin 16 times, either a real one or [virtually](#), and use the sheet [linked here](#) to report how many "heads" you flipped. Repeat this process (reporting your results each time) for **at least three sets of 16 flips**. We will look at the class's data together.